

HHS3190M
Human Physiology and Functional Anatomy
for Rehabilitation Services
HHS3190MJ

復康服務之人體生理學及功能性解剖學

Physiology L7: Endocrine System
生理學 L7: 內分泌系統

(Sem 1 AY 26-27 26-27學年第1學期)

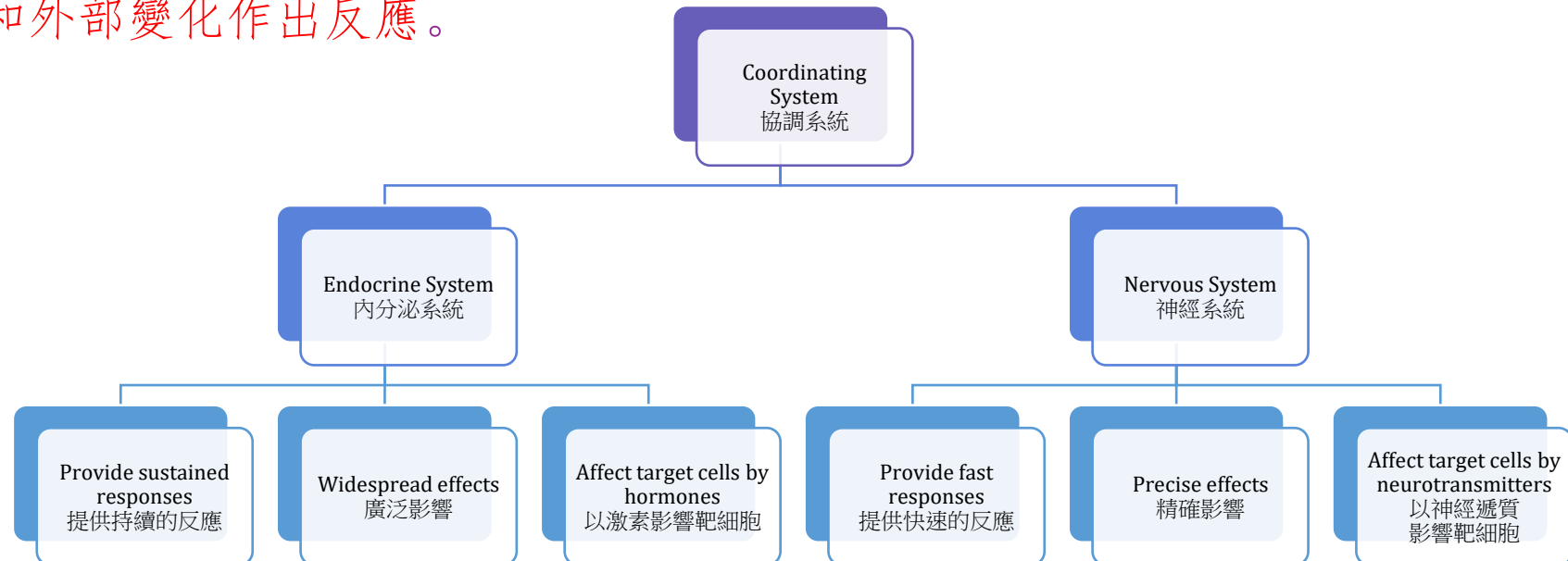
Outline 大綱

- Endocrine system and major endocrine glands
內分泌系統和主要內分泌腺體
- Hormone chemical structures
激素化學結構
- Mechanisms of hormone action
激素作用機制
- Regulation of hormone secretion
激素分泌的調節
- Examples of hormone imbalance
激素失衡的例子
(Self-Study 自學)

Overview of endocrine system 內分泌系統概述

- The **endocrine system** and **nervous system** are two major parts of human's coordinating system to integrate activities of all cells and responses to both internal and external changes.

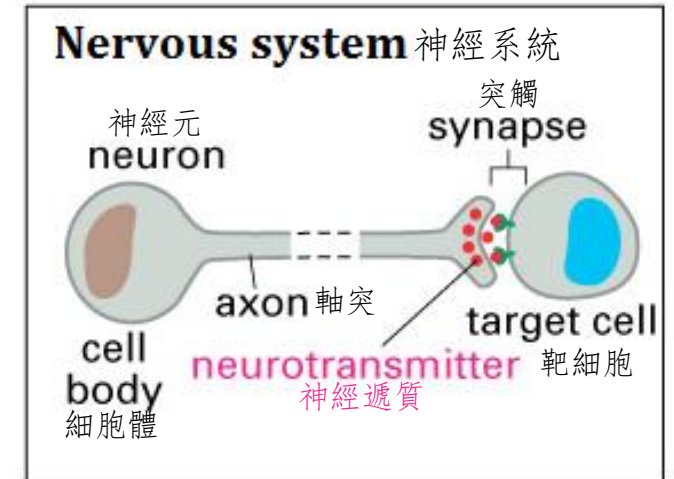
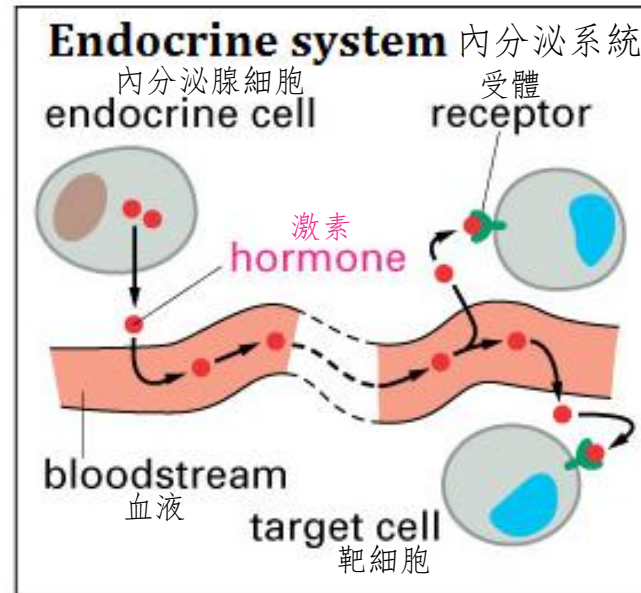
內分泌系統和神經系統是人體協調系統的兩個主要部分，用於整合所有細胞的活動以及對內部和外部變化作出反應。



Overview of endocrine system 內分泌系統概述

- Endocrine system 內分泌系統:

- A cell (**Endocrine gland cell**) secretes a chemical. 細胞(內分泌腺細胞)分泌一種化學物質。
- The chemical (**Hormone**) travel through the **blood**. 化學物質(激素)通過血液運輸。
- Another cell (**Target cell**) is stimulated to **change activities**. 另一個細胞(靶細胞)受到刺激改變活動。



Overview of endocrine system 內分泌系統概述

- Different effects of hormones 激素的不同作用:
 - Reproduction (E.g. estrogen and testosterone)
生殖(例如雌激素和睪固酮)
 - Growth and development (E.g. growth hormone)
生長發育(例如生長激素)
 - Balance of ions, water, and nutrients (E.g. insulin)
離子、水和營養物質的平衡(例如胰島素)
 - Regulation of cellular metabolism (E.g. epinephrine)
細胞代謝的調節(例如腎上腺素)
 - Regulation of body defenses (E.g. Thymosin)
身體防禦的調節(例如胸腺素)

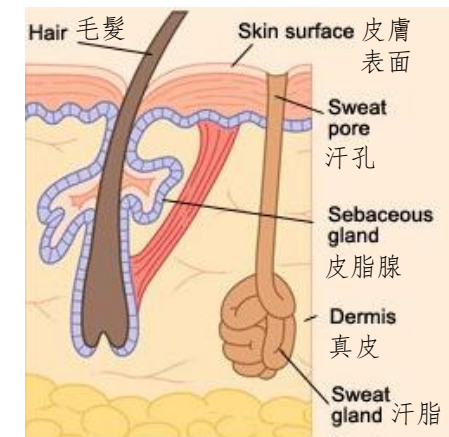
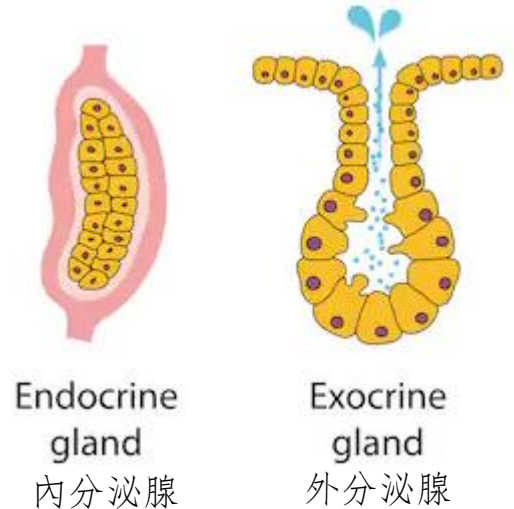
Overview of endocrine system

內分泌系統概述

- “Glands” in human body can be divided into **endocrine glands** and **exocrine glands**.

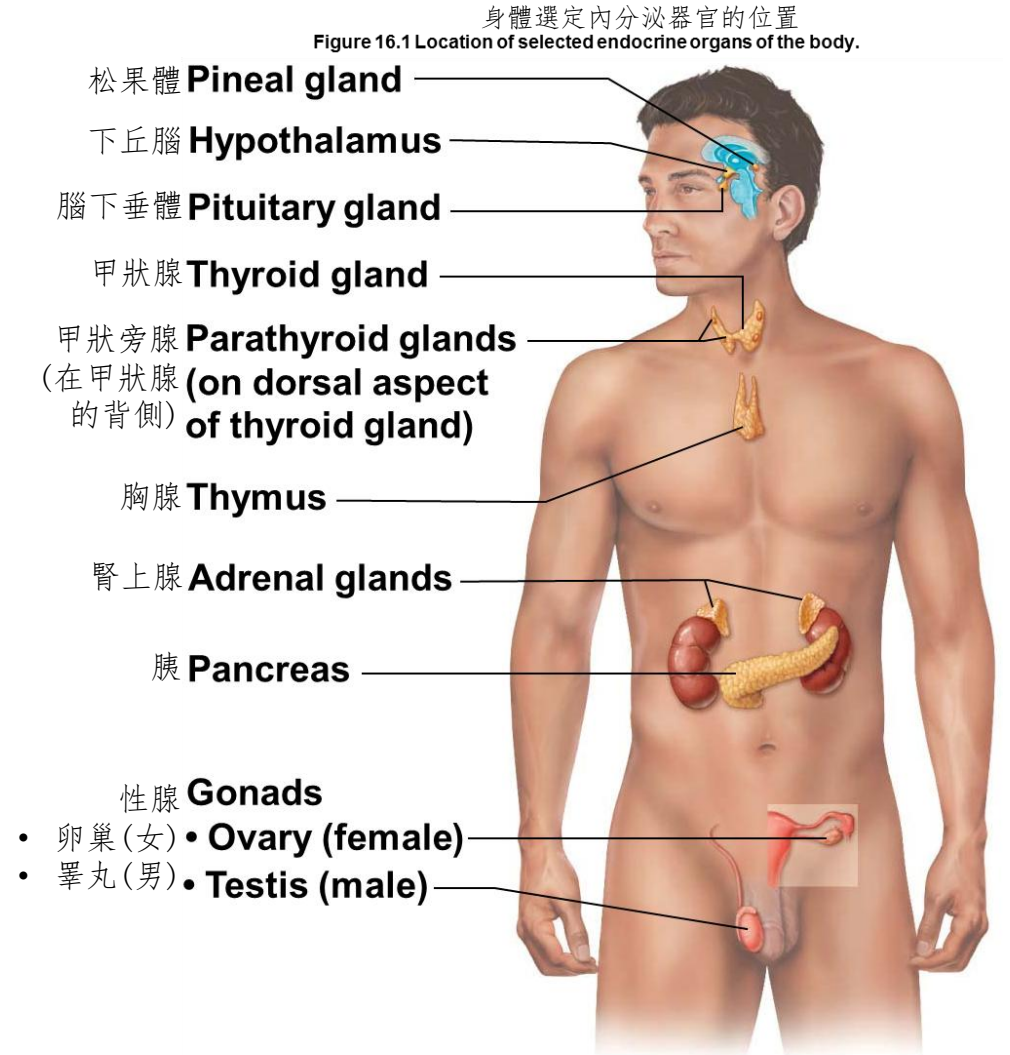
人體內的「腺體」可分為**內分泌腺**和**外分泌腺**。

- Endocrine glands: **ductless** glands that secrete **hormones** into **blood** 內分泌腺：將**激素**分泌到**血液**中的**無管**腺
- Exocrine glands: glands that secrete **nonhormone chemicals** to a **surface** through **ducts** 外分泌腺：通過**導管**向**表面**分泌**非激素化學物質**的腺體
- Sweat gland, skin sebaceous gland, and gastric gland are examples of exocrine glands. 汗腺、皮膚皮脂腺和胃腺是外分泌腺的例子。
- Only endocrine glands belong to “endocrine system”. 只有內分泌腺屬於「內分泌系統」。



Major endocrine glands 主要內分泌腺體

Endocrine gland 內分泌腺	Major hormone (s) and function(s) 主要激素和功能
Pineal gland 松果體	Melatonin 褪黑激素 • Regulating “biological clock” 調節「生物鐘」
Thyroid gland 甲狀腺	Thyroxine (T4) 甲狀腺素 • Increase rate of metabolism 增加新陳代謝率
Parathyroid gland 甲狀旁腺	Parathyroid hormone (PTH) 甲狀旁腺激素 • Balance of calcium ions 鈣離子平衡
Adrenal gland 腎上腺	Epinephrine (Epi) 腎上腺素 • Enhance “fight-or-flight” responses 增強「戰鬥或逃跑」反應 Cortisol 皮質醇 • Change metabolism to prolong stress responses 改變新陳代謝以延長壓力反應 Aldosterone 醛固酮 • Balance of sodium and potassium ions 鈉離子和鉀離子的平衡
Pancreas 胰	Insulin 胰島素 • Decrease blood glucose 降低血糖 Glucagon 胰高血糖素 • Increase blood glucose 增加血糖
Ovary 卵巢	Estrogen 雌激素 (Discuss in “Reproductive system”) (在“生殖系統”中討論)
Testis 睪丸	Testosterone 睪固酮 (Discuss in “Reproductive system”) (在“生殖系統”中討論)



Endocrine gland 內分泌腺	Major hormone(s) 主要激素	Major function(s) 主要功能
Pituitary gland - Posterior lobe 腦下垂體 - 後葉	Oxytocin 催產素	Stimulate contraction of uterus; 刺激子宮收縮; Milk “let-down” reflex 射乳反射
	Antidiuretic hormone (ADH) 抗利尿激素	Promote water reabsorption by kidneys 促進腎臟對水分的重吸收
Pituitary gland - Anterior lobe 腦下垂體 - 前葉	Growth hormone (GH) 生長激素	Stimulate growth (especially bones and muscles) and metabolism 刺激生長(尤其是骨骼和肌肉)和新陳代謝
	Prolactin (PRL) 催乳素	Stimulate milk production 刺激產乳
	Adrenocorticotrophic hormone (ACTH) 促腎上腺皮質激素	Stimulate adrenal gland to secrete cortisol 刺激腎上腺分泌皮質醇
Pituitary gland - Anterior lobe 腦下垂體 - 前葉	Follicle stimulating hormone (FSH) 促卵泡激素	Stimulate production of sperm and ova 刺激精子和卵子的產生
	Luteinizing hormone (LH) 黃體生成素	Stimulate sex hormone secretion; 刺激性激素分泌; Stimulate ovulation 刺激排卵
	Thyroid stimulating hormone (TSH) 促甲狀腺激素	Stimulate thyroid to secrete thyroid hormones (Thyroxine) 刺激甲狀腺分泌甲狀腺激素(甲狀腺素)

Hormone Chemical Structure 激素化學結構

- Chemical structure of a hormone determines its solubility in water. This is related to:
激素的化學結構決定了其在水中的溶解度，這與以下有關：
 - How is it transported in blood? 它是如何在血液中運輸的？
 - How does it affect its target cells? 它如何影響其靶細胞？

- Two classes: 兩個類別：

- (1) **Protein or amino acid based** 基於蛋白質或氨基酸

Most hormones are proteins or amino acid based. (E.g. LH, Epi, ADH)
大多數激素是基於蛋白質或氨基酸的。(例如LH、Epi、ADH)

These are usually **water soluble** and **cannot cross cell membrane**.
它們通常是水溶性的，不能穿過細胞膜。

- (2) **Steroids** 類固醇

Some hormones are made from cholesterol. (E.g. aldosterone, estrogen, testosterone)
有些激素是由膽固醇製成的。(例如醛固酮、雌激素、睪固酮)

These are all **lipid soluble** and **can cross cell membrane**.
這些都是脂溶性的，可以穿過細胞膜。

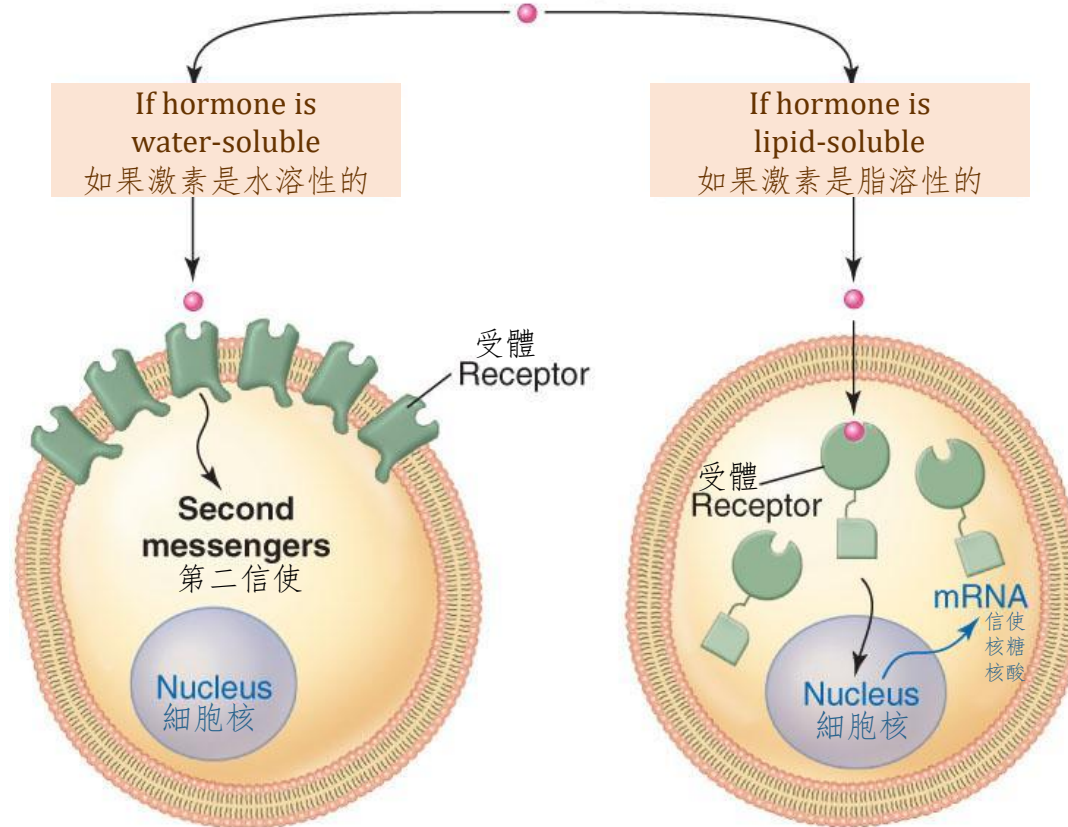
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Hormone is carried by blood

to target cells

激素由血液運輸到靶細胞

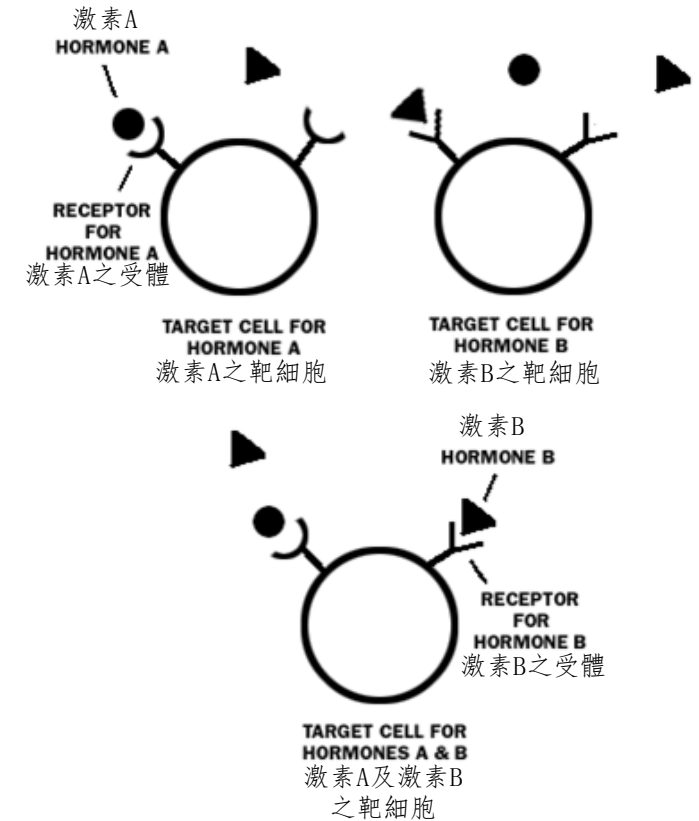
**Protein or
amino acid
based
hormones**
蛋白質或
氨基酸激素



**Steroid
hormones**
類固醇激素

Mechanisms of Hormone Action 激素作用機制

- Hormones are carried by blood to all tissues.
However, not all cells are affected by 1 hormone.
血液運輸激素到全身組織，但並非所有細胞都受到1種激素的影響。
- The cells, that are affected, are “**target cells**” of a certain hormone.
受影響的細胞被稱為某種激素的“**靶細胞**”。
 - These cells have specific **receptor proteins** in cytoplasm or in cell membrane.
這些細胞在細胞質或細胞膜中具有**特定的受體蛋白**。
 - **Hormone molecules bind to receptor proteins** and trigger a series of changes of these cells.
激素分子與受體蛋白結合並引發這些細胞的一系列變化。



Mechanisms of Hormone Action 激素作用機制

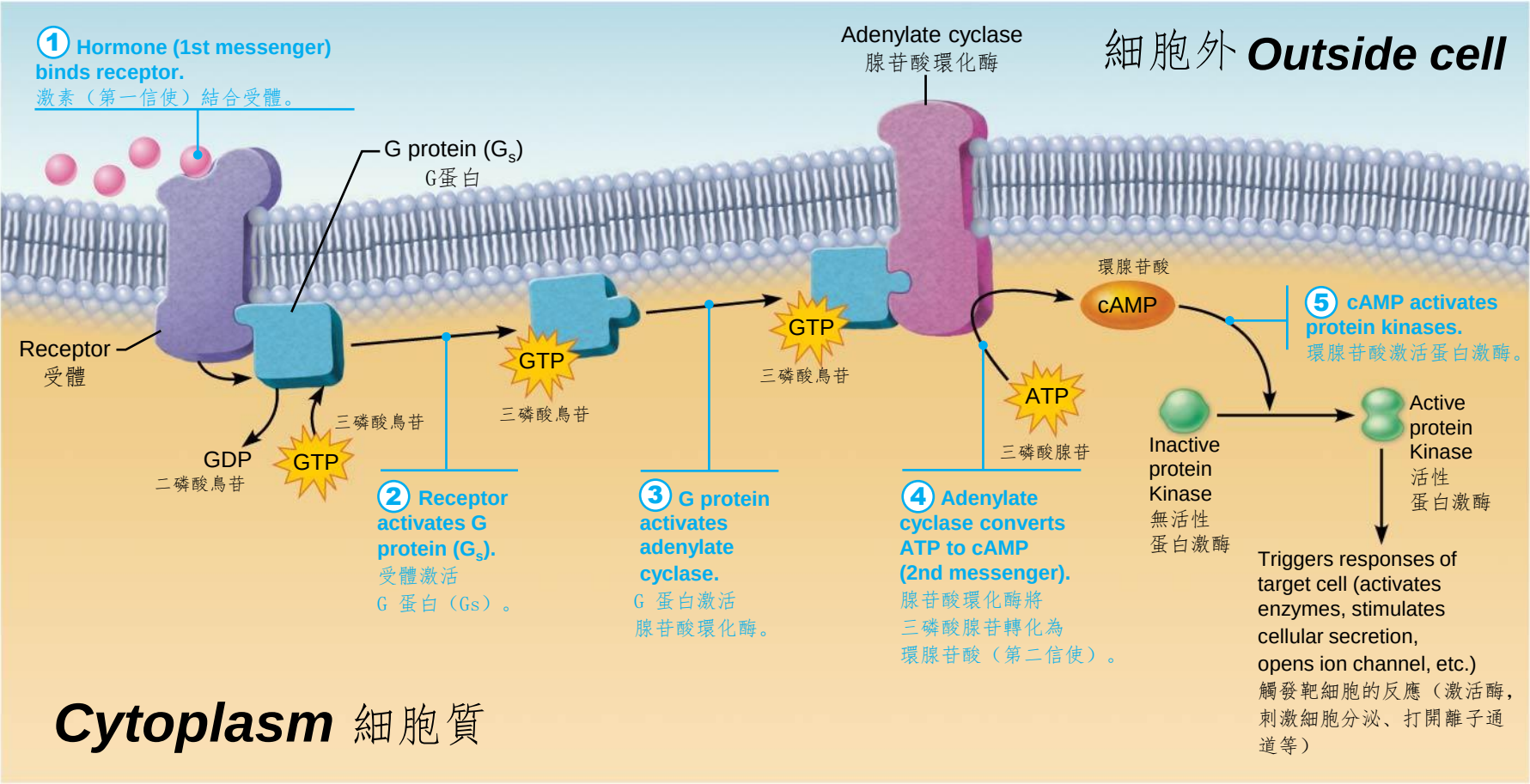
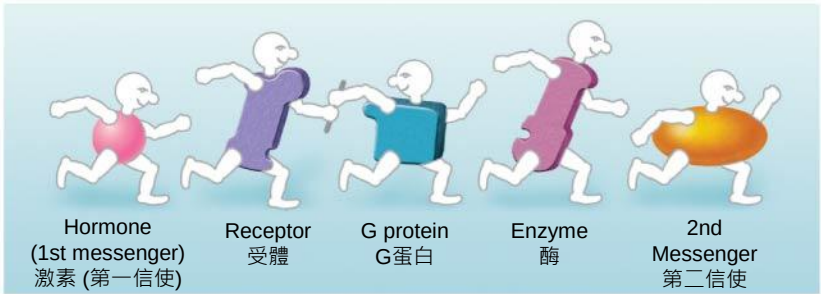
- Mechanisms of **water soluble hormones**:

水溶性激素的機制：

- (1) Bind to “**plasma membrane receptors**” on cell membrane
與細胞膜上的「細胞表面受體」結合
- (2) Trigger increase of different “**second messengers**” inside the cell
觸發細胞內部不同的「第二信使」的增加
 - Examples of common second messengers include **cAMP**, **calcium ions**, cGMP, etc.
常見的第二信使的例子包括環腺苷酸、鈣離子、環磷酸鳥苷等。
- (3) Second messengers **activate or inactivate enzymes** inside cytoplasm.
第二信使激活或失活細胞質內的酶。
- (4) **Cellular activities are changed!**
細胞活動改變！

Cyclic AMP second-messenger mechanism of water-soluble hormones. 水溶性激素的環腺苷酸第二信使機制

G protein signaling mechanisms are like a molecular relay race.
G 蛋白信號傳導機制就像一場分子接力競賽。



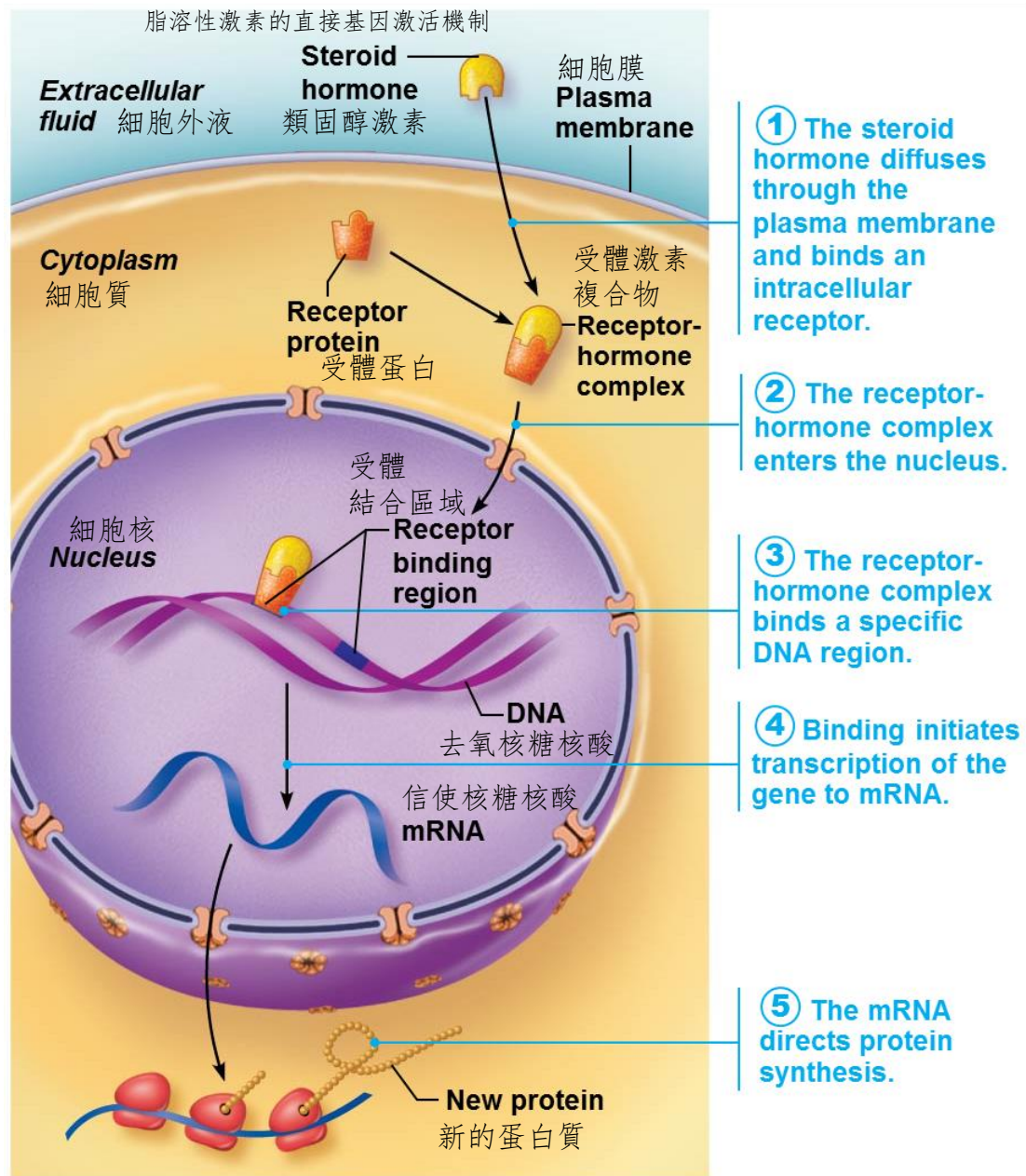
Mechanisms of Hormone Action 激素作用機制

- Mechanisms of **lipid soluble hormones**:

脂溶性激素的機制：

- (1) Bind to “**intracellular receptors**” in cytoplasm or in nucleus
與細胞質或細胞核中的「**細胞內受體**」結合
- (2) **Hormone-receptor complex** binds to a **specific DNA region**
激素受體複合物與特定去氧核糖核酸(DNA)區域結合
- (3) A specific **mRNA** is produced from certain region of DNA.
特定的**信使核糖核酸(mRNA)**由核糖核酸(DNA)的某個區域產生。
- (4) The mRNA leaves nucleus and attaches to **ribosomes on rough ER**.
信使核糖核酸(mRNA)離開細胞核並附著在**粗面內質網上的核糖體**上。
- (5) **New proteins** are produced to **change cellular activities**.
產生**新的蛋白質**來**改變細胞活動**。

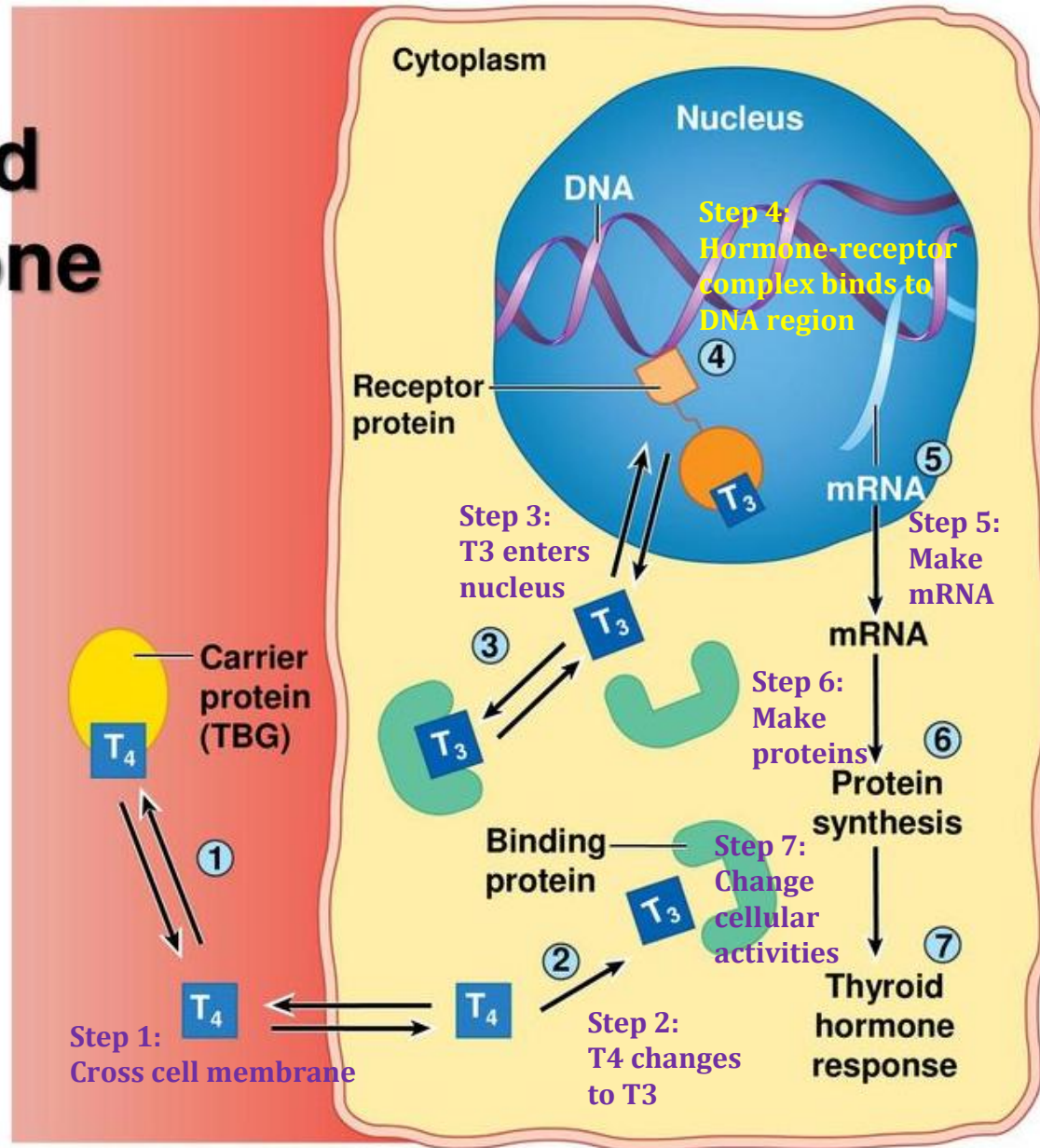
Figure 16.3 Direct gene activation mechanism of lipid-soluble hormones.



- ① 類固醇激素擴散通過細胞膜並結合細胞內受體。
- ② 受體激素複合物進入細胞核。
- ③ 受體激素複合物結合特定的去氧核糖核酸 (DNA) 區域。
- ④ 結合觸發基因轉錄為信使核糖核酸 (mRNA)。
- ⑤ 信使核糖核酸 (mRNA) 控制蛋白質合成。

Thyroid Hormone

甲狀腺激素



Thyroxine (T4) and Triiodothyronine (T3) are called “thyroid hormones”.

They are amino acid-based but they are water-insoluble!

甲狀腺素 (T4) 和三碘甲狀腺素 (T3) 被稱為“甲狀腺激素”。

它們以氨基酸為基礎，但不是水溶性！

① 甲狀腺素 (T4) 透過載體蛋白運輸到靶細胞，脫離載體蛋白並擴散通過細胞膜。

② 在細胞質內 T4 被轉換為三碘甲狀腺素 (T3)。

③ T3 進入細胞核。

④ 在細胞核內，T3 結合受體蛋白形成受體激素複合物，然後結合特定的去氧核糖核酸 (DNA) 區域。

⑤ 結合觸發基因轉錄為信使核糖核酸 (mRNA)。

⑥ 信使核糖核酸 (mRNA) 控制蛋白質合成。

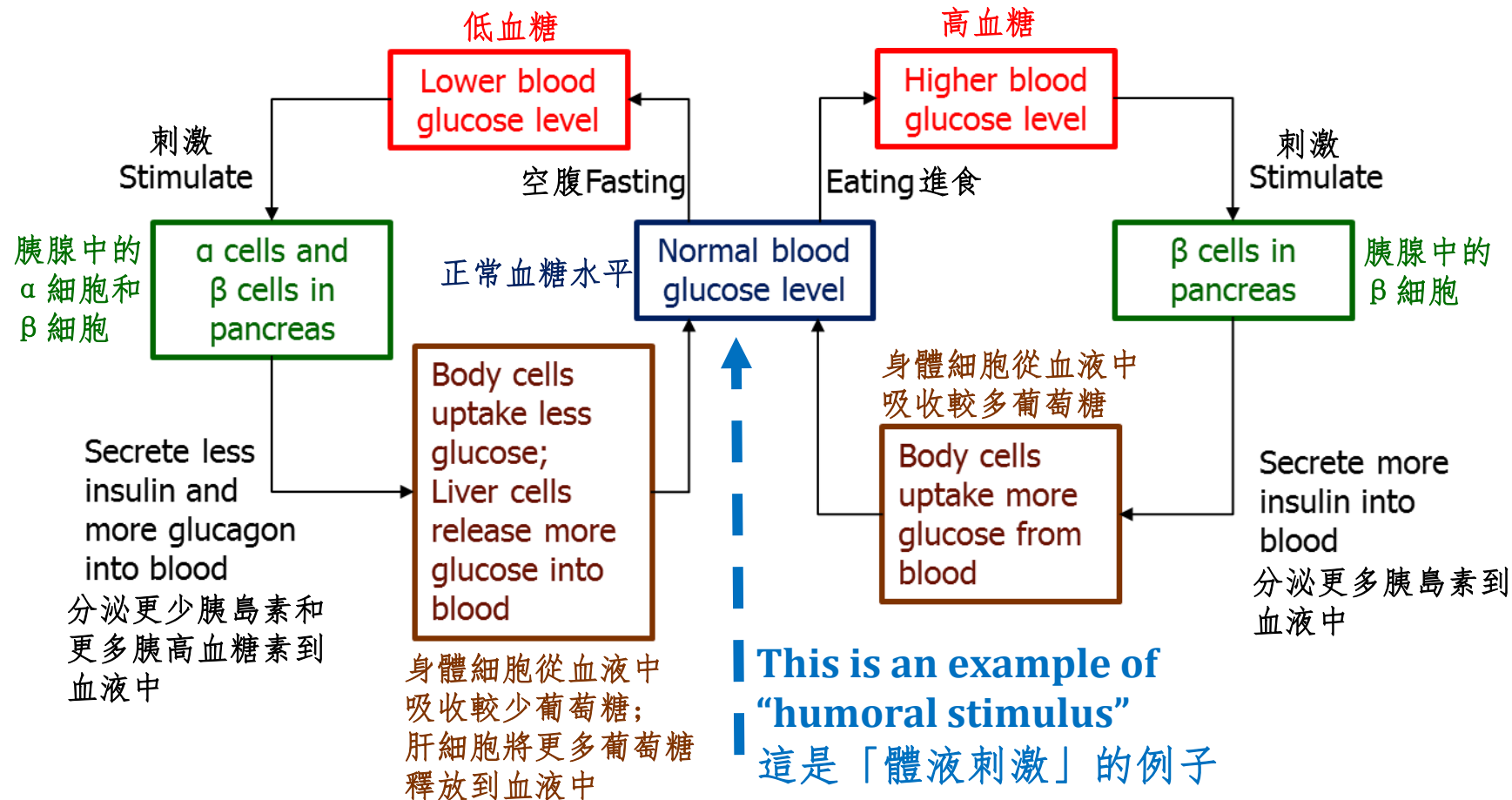
⑦ 新的蛋白質改變細胞活動，出現甲狀腺激素反應。

Regulation of Hormone Secretion 激素分泌的調節

- The synthesis and release of most hormones are regulated by some type of **negative feedback mechanism**. 大多數激素的合成和釋放受到某種**負反饋機制**的調節。
- 3 types of stimuli trigger endocrine glands to release more hormones.
3 類刺激會觸發內分泌腺釋放更多激素。
 - (1) **Humoral stimuli** 體液刺激:
concentrations of certain molecules (E.g. glucose) or ions (E.g. Calcium ions) in blood
血液中某些分子(例如葡萄糖)或離子(例如鈣離子)的濃度
 - “**Low Na⁺ or high K⁺ blood levels** 低血鈉或高血鉀”
 - ➔ Adrenal gland secretes **more aldosterone** 腎上腺分泌更多的醛固酮
 - ➔ Increase Na⁺ and decrease K⁺ 增鈉及減鉀
 - (2) **Neural stimuli** 神經刺激: ANS controls endocrine glands 自主神經系統控制內分泌腺
 - “**Sympathetic nervous system** 交感神經系統”
 - ➔ Adrenal gland secretes **more epinephrine (Epi)** 腎上腺分泌較多的腎上腺素
 - ➔ Enhance “Fight-or-flight” responses 增強「戰鬥或逃跑」反應
 - (3) **Hormonal stimuli** 激素刺激:
concentrations of other hormones in blood 血液中其他激素的濃度

Regulation of Hormone Secretion 激素分泌的調節

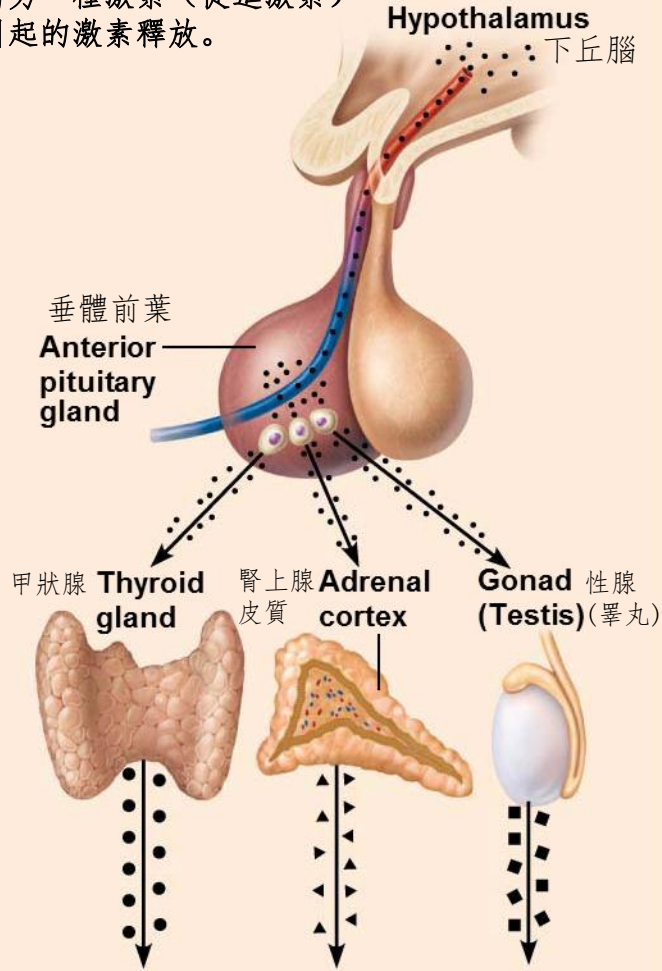
- Pancreas and its endocrine function 胰及其內分泌功能



Hormonal stimuli 激素刺激

Hormone release caused by another hormone (a tropic hormone).

由另一種激素（促進激素）引起的激素釋放。

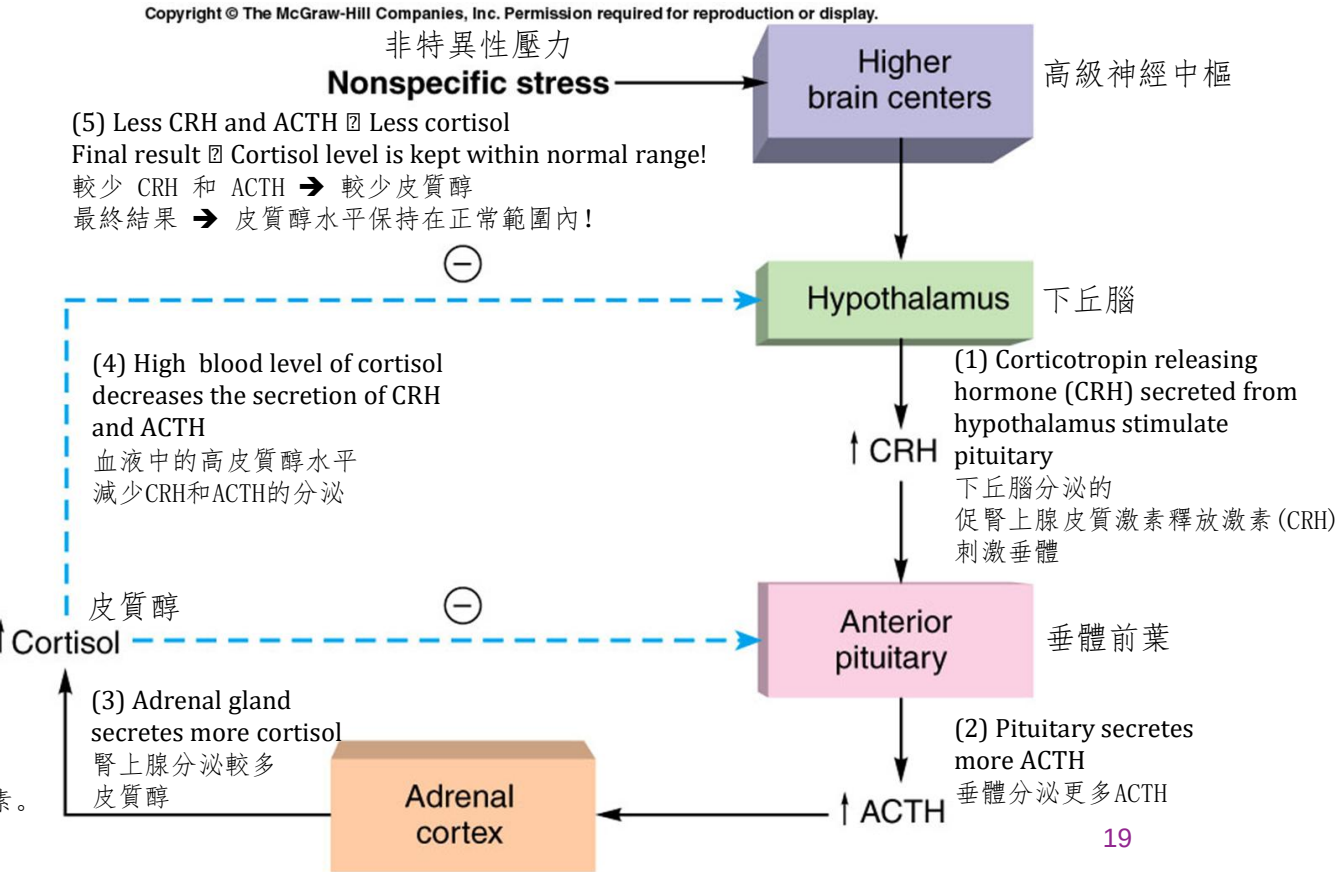


Stimulus: Hormones from hypothalamus.
Response: Anterior pituitary gland secretes hormones that stimulate other endocrine glands to secrete hormones.

刺激：來自下丘腦的激素
反應：垂體前葉分泌激素，刺激其他內分泌腺分泌其他激素。

Regulation of Hormone Secretion 激素分泌的調節

Hypothalamus-pituitary-adrenal axis 下丘腦-垂體-腎上腺軸

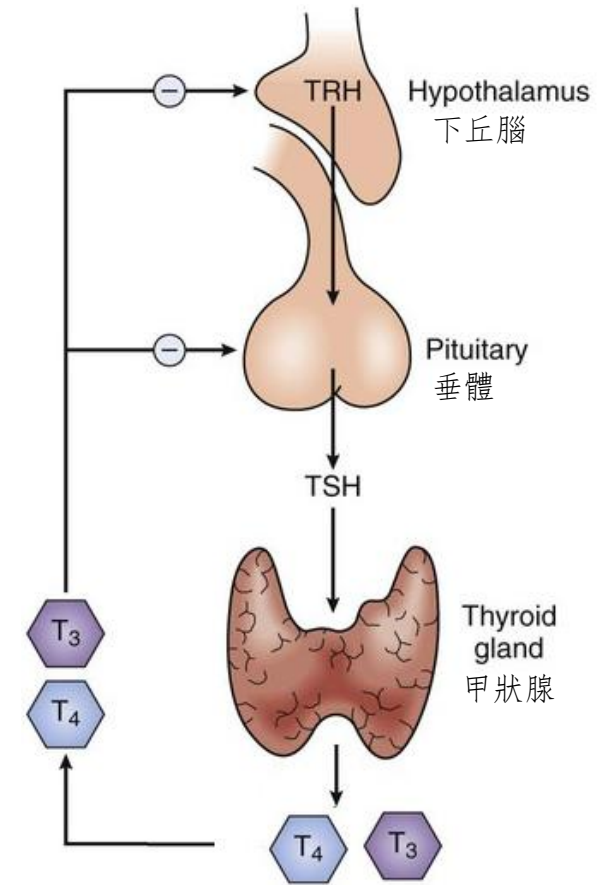


Regulation of Hormone Secretion 激素分泌的調節

• Hypothalamus-pituitary-thyroid axis

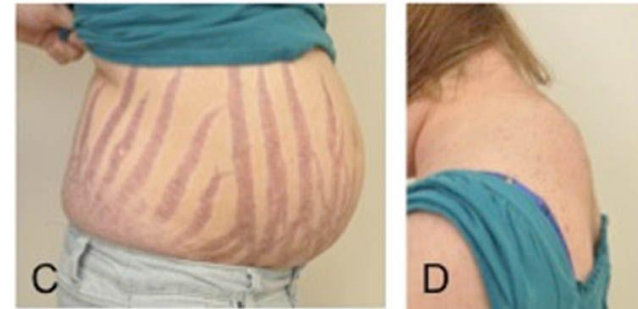
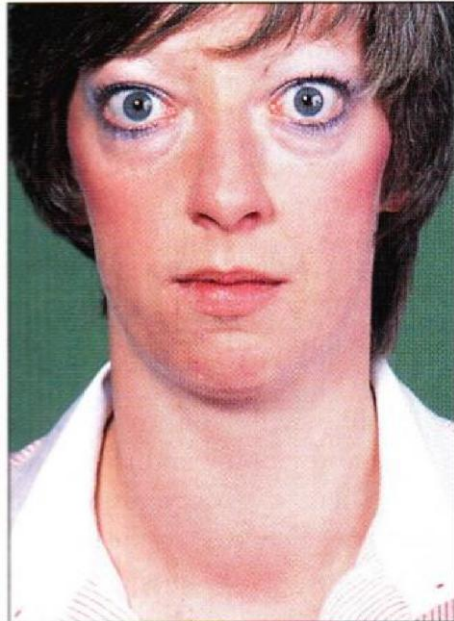
下丘腦-垂體-甲狀腺軸

- (1) Thyrotropin releasing hormone (TRH) secreted from hypothalamus stimulate pituitary
下丘腦分泌的促甲狀腺素釋放激素 (TRH) 刺激垂體
- (2) Pituitary secretes more TSH
垂體分泌多促甲狀腺素 (TSH)
- (3) Thyroid gland secretes more T3 and T4
甲狀腺分泌較多 T3 和 T4
- (4) High blood levels of T3 and T4 decrease the secretion of TRH and TSH
血液中 T3 和 T4 的高水平減少 TRH 和 TSH 的分泌
- (5) T3 and T4 secretions decrease and blood levels of T3 and T4 are kept within normal range
T3 和 T4 分泌減少，T3 和 T4 的血液水平保持在正常範圍內



Examples of Hormone Imbalance 激素失衡的例子 (Self-Study自學)

Condition 狀況	Hormone imbalance 激素失衡	Features 特徵
Diabetes mellitus (DM) 糖尿病	Lack insulin OR Cells do NOT response to insulin 缺乏胰島素或 細胞對胰島素沒有反應	High blood glucose level 高血糖 Excess urine 多尿 Excess thirst 過度口渴 Excess hunger 過度飢餓
Hyperthyroidism 甲狀腺功能亢進	Too much thyroid hormone 甲狀腺激素過多	Enlarged thyroid (goiter) 甲狀腺腫大(甲狀腺腫) Rapid metabolism and weight loss 新陳代謝快和減重 Rapid heartbeat 心跳快 Sweating, nervousness, feeling hot 出汗、精神緊張、感覺熱 Bulging eyes 眼睛凸出
Hypothyroidism 甲狀腺功能減退	Lack thyroid hormone 缺乏甲狀腺激素	Enlarged thyroid (goiter) 甲狀腺腫大(甲狀腺腫) Slow metabolism and weight gain 新陳代謝慢和增重 Mental sluggishness, feeling cold 精神遲鈍，感覺冷 Puffy eyes 眼睛浮腫
Cushing's syndrome 庫欣綜合徵	Too much cortisol (or other chemicals with similar structure) 皮質醇過多 (或其他具有類似結構 的化學物質)	High blood glucose level 高血糖 High blood pressure 高血壓 Central obesity with thin limbs 中央肥胖而四肢瘦削 Moon face and buffalo hump 月亮臉和水牛肩
Acromegaly 肢端肥大症	Excess growth hormone in adult 成人生長激素過量	Overgrowth of bones in hands, feet, and face 手、腳和臉的骨骼過度生長



L7 Revision Exercise 複習練習

- Screen the QR code below to complete all questions in this revision exercise.
掃描下面的二維碼以完成此複習練習中的所有問題。



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